

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (E&C) (Credit System) Degree Examinations

January - 2022

19ECM102 – VLSI CAD

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full questions** choosing **Two full questions** from **Unit- I & Unit- II** each and **One full question** from **Unit – III**.

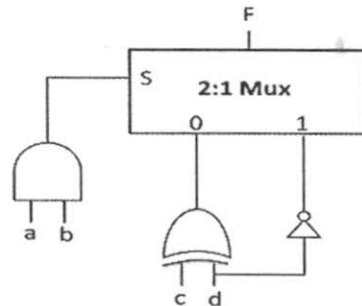
Unit – I

Marks BT* CO* PO*

1. a) Show the cubic representation of one variable, two variable and three variable Boolean function. Simplify the Boolean Expression $Y(a,b,c) = \bar{a}b\bar{c} + \bar{a}bc + ab\bar{c} + abc$ and draw the cubic representation for the simplified expression.

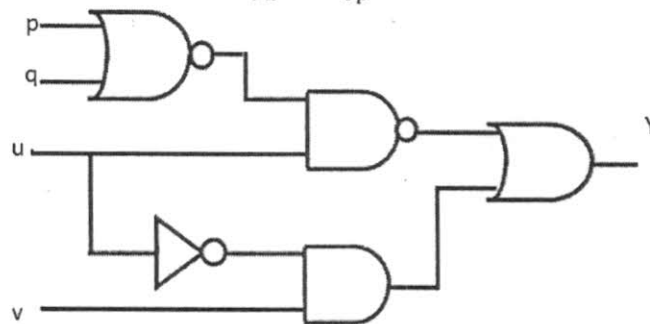
10	L*3	1	1
----	-----	---	---
- b) Realize the function $F(a,b,c) = \sum m(3,4,5,6,7)$ using
 - i) Binary Tree
 - ii) Reduced Binary Tree

10	L3	2	1
----	----	---	---
2. a) For the logic diagram shown in Figure 2.(a), write the logic expression and also determine the following quantification operators. i. $(\forall cd)f(a,b)$
ii. $(\exists ab)f(c,d)$

**Figure 2.(a)**

10 L3 1 1

- b) For the logic diagram shown in Figure 2.(b), write the simplified logic expression and find (i) $\frac{\partial Y}{\partial u}$ (ii) $\frac{\partial Y}{\partial p}$

**Figure 2.(b)**

10 L3 1 1

3. a) State Shannon's expansion theorem. Implement the following function using 4 to 1 multiplexer and draw the logic diagram.
 $f(x_1, x_2, x_3) = \bar{x}_1 * \bar{x}_3 + x_1x_2 + x_1x_3$

10 L3 2 1

P.T.O.

b) With an example explain the following terms.

- (i) Boolean Product
- (ii) Negative unate function
- (iii) Positive unate function
- (iv) Binate function
- (v) Algebraic Product

10 L2 1 1

Unit – II

4. a) For the given functions f_1 and f_2 , obtain the prime implicants using K-map and sketch the minimized two level circuit.

$$f_1(a, b, c) = \pi M(0, 1, 3, 4, 5)$$

$$f_2(a, b, c) = \pi M(0, 4, 5, 6, 7)$$

10 L3 3 1

b) Determine the kernels and co-kernels of the function Y, where $Y = ABDE + ABDG + EH$.

10 L2 4 1

5. a) Define kernel. Explain level – 0 and level – n kernel. Given $Y = AE + AF + BCE + BCF + BDE + BDF + G$. Determine level-0, level-1 and level-2 kernels for this function

10 L3 4 1

b) For the SOP expression, $y = abdf + caf + \bar{a}\bar{e}f + \bar{b}\bar{d}\bar{f} + \bar{b}e\bar{f}$

- (i) Determine the factored expression
- (ii) Determine the network graph for SOP and factored expression
- (iii) Draw the two level and multilevel logic circuit.

10 L3 4 1

6. a) Give the drawback of heuristic two level circuit minimization method. Explain in detail how ESPRESSO overcomes this limitation.

10 L2 3 1

b) Briefly explain the terms controllability and observability with respect to Don't cares.

4 L2 4 1

c) Explain the significance of Don't cares in minimizing the cost of switching function in logic design.

6 L2 4 1

Unit – III

7. a) Explain Simulated Annealing algorithm with respect to placement.

10 L2 5 1

b) Explain the two approaches to solve the global routing problem

10 L2 5 1

8. a) What is technology mapping? How the circuit components are mapped to a set of logic cells using technology mapping. Explain with an example.

10 L2 5 1

b) Explain in detail three different levels of placement.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
